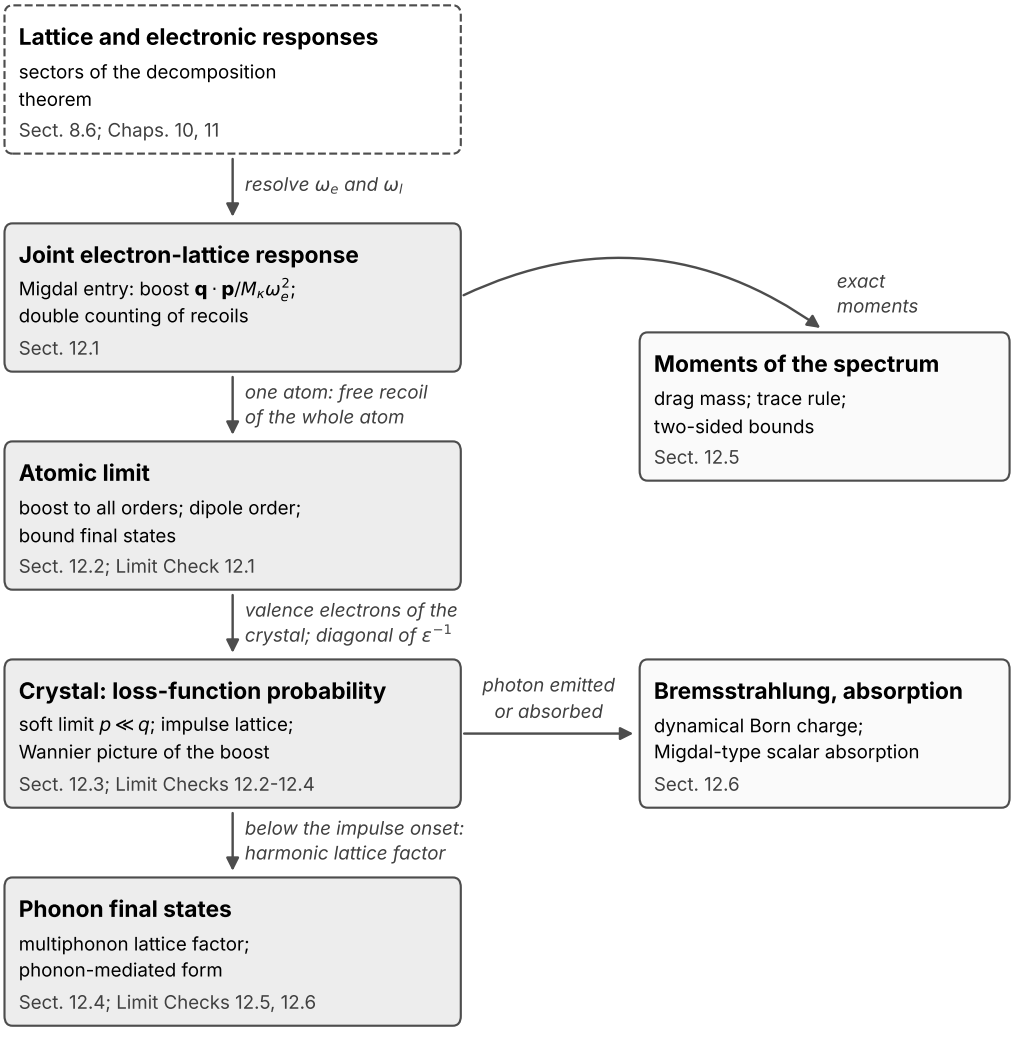
Lattice and electronic responses
sectors of the decomposition theorem
Sect. 8.6; Chaps. 10, 11
resolve ω_e and ω_I
Joint electron-lattice response
Migdal entry: boost $\mathbf{q}\cdot\mathbf{p}/M_\kappa\omega_e^2$; double counting of recoils
Sect. 12.1
exact moments
Moments of the spectrum
drag mass; trace rule; two-sided bounds
Sect. 12.5
one atom: free recoil of the whole atom
Atomic limit
boost to all orders; dipole order; bound final states
Sect. 12.2; Limit Check 12.1
valence electrons of the crystal; diagonal of ε^{-1}
Crystal: loss-function probability
soft limit $p \ll q$; impulse lattice; Wannier picture of the boost
Sect. 12.3; Limit Checks 12.2-12.4
photon emitted or absorbed
Bremsstrahlung, absorption
dynamical Born charge; Migdal-type scalar absorption
Sect. 12.6
below the impulse onset: harmonic lattice factor
Phonon final states
multiphonon lattice factor; phonon-mediated form
Sect. 12.4; Limit Checks 12.5, 12.6